

Zixuan Guo

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SUMMARY

Audio AI & Music Technology researcher-developer (M.M., NYU) building machine learning systems for music/audio and interactive media. Interests: controllable music generation, robust machine listening, audio representation learning, and human-centered AI tools for creative workflows.

EDUCATION

New York University

Master of Music (M.M.) in Music Technology

New York, NY

Sep 2024 – May 2026

GPA: 3.98/4.0 | **Coursework:** Deep Learning for Media, Machine Listening, Digital Signal Processing, Music Information Retrieval, C++ Audio Application Development

Jilin University

Bachelor of Science (B.S.) in Computer Science and Technology

Jilin, China

Sep 2019 – Jun 2024

GPA: 3.41/4.0 | **Coursework:** Object-Oriented Programming, Software Engineering, Algorithms & Data Structures, Operating Systems, Machine Learning, Linux Systems

PUBLICATIONS

Guo, Z., Zhong, J., & Chen, S. (2026). Noise-robust instrument classification under real-world SNR conditions. Preprint submitted to SSRN.

AWARDS

1. **Academic Achievement Recognition**, Graduate Music Technology Program, NYU Steinhardt May 2026
2. **Third-Class Scholarship**, Department of Computer Science and Technology, Jilin University Sep 2020

WORK EXPERIENCE

Hazel Investment

Data Mining & Analytics Intern

Shenzhen, China

Jan 2024 – Feb 2024

- Supported quantitative analysis and analytics workflows on financial datasets using Python.
- Conducted exploratory data analysis (EDA), preprocessing, visualization, and automated reporting for market intelligence tasks.
- Collaborated with analysts on data-oriented research and reporting projects.

University of Washington (UW)

Research Assistant (DeepSqueak)

Seattle, WA, USA

May 2025 – Nov 2025

- Modernized a legacy spectrogram-based acoustic event detector by migrating from YOLOv2 to anchor-free YOLOX, reducing manual bounding-box editing and improving contour localization.
- Engineered scalable MATLAB preprocessing for multi-channel spectrogram generation, dynamic noise augmentation, and hard-negative mining.
- Improved low-SNR generalization via targeted data refinement and augmentation strategies; summarized methods and results in a research report.
- GitHub Link

RESEARCH EXPERIENCE

Robust Acoustic Scene Classification Research (DCASE / ConvMixer-RSC)

Spring 2026

- Reproduced and extended Truchan et al.'s ConvMixer+RSC baseline for DCASE 2024 Task 1 under strict constraints (<128 KB params, <30 MMAC/clip) and low-label regimes (5%–100%).
- Built a deterministic benchmarking pipeline (data splits, augmentation, checkpointing, per-class/per-device metrics, MAC/parameter profiling) for fair ablations.
- Evaluated a learnable EfficientLEAF frontend as a drop-in replacement for mel; observed no gains at 5% as filterbank parameters stayed near mel initialization (adaptation concentrated in per-channel gain).
- Added Efficient Channel Attention (ECA) to ConvMixer blocks and tuned kernel size; $k=7$ improved 5% macro accuracy from 39.29% to 39.69% while reducing compute from ~ 29.4 to ~ 26.5 MMACs by using an 8-block backbone.
- GitHub Link

Audio Backend Developer | XRcising (Mixed-Reality Fitness Platform)

Spring 2026

- Proposed an audio feature extraction and event generation pipeline converting user-uploaded music into time-aligned XR interaction targets for Meta Quest.
- Designed a prototype MIR pipeline integrating beat tracking, spectral features, and source separation (nussl) for spatial interaction mapping.

Generative AI Developer | Master's Thesis: Controllable Symbolic Music Generation

May 2026

- Proposed a decoupled controllable symbolic music generation framework: probabilistic proposal (MuseCoco autoregressive Transformer) followed by deterministic, constraint-based symbolic post-processing (control layer + cleanup + rule layer).
- Implemented three interaction modes in a unified pipeline: New/Continue generation + post-hoc control, and Transformation via attribute-regularized VAE (AR-VAE) latent-space editing followed by cleanup.
- Built a JUCE DAW plugin connected to a backend inference service, enabling iterative, non-destructive refinement inside standard production workflows.
- Evaluated controllability via ablations and objective constraints; improved New-mode success from 48% (MuseCoco-only) to 76% with controls (+28 pp) and ran a 7-participant DAW user study (usability mean 1.81/5; lower is better).
- GitHub Link

Pitch Tracking in Non-Ideal Conditions

Fall 2025

- Built a reproducible pitch-tracking benchmark on MedleyDB-Pitch to quantify robustness under real-world perturbations: tanh soft-clipping distortion ($g=2/5/7.5$), additive noise (room/street/crowd at 15/5 dB SNR), and global detuning ($\pm 25/\pm 50$ cents) with annotation-consistent pitch shifts.
- Implemented the data loading + evaluation pipeline for Librosa pYIN, CREPE, and Spotify Basic Pitch; batch-ran experiments and generated summary visualizations/tables.
- Standardized heterogeneous model outputs by normalizing predictions onto the ground-truth timestamp grid (nearest-neighbor alignment + voiced-only linear interpolation) to enable fair metric comparisons.
- Computed MIR evaluation metrics (OA, RPA, RCA, VR) and analyzed failure modes; found Basic Pitch most robust across noise/distortion/detuning, while CREPE was highly accurate on clean audio but fragile under perturbations.
- GitHub Link

Noise-Robust Musical Instrument Classification

Spring 2025

- Built a noise-robust instrument classifier targeting real-world low-SNR recordings by combining LAION-CLAP (LionCLAP) embeddings with noise-aware curriculum training.
- Constructed a noisy training set by mixing MUSAN noise into Medley-solos-DB at 20/10/5/0 dB SNR and trained models progressively from high to low SNR.
- Implemented and compared TCN vs. lightweight Transformer sequence classifiers (freeze vs. fine-tune CLAP); achieved 95.93% (clean) / 81.28% (SNR=-5) accuracy with TCN (no-freeze).
- Added a confidence-triggered harmonic-enhanced fallback and negative (pure-noise) samples to reduce extreme-noise misclassification and explicitly reject non-instrument inputs.
- GitHub Link

Machine Learning Developer | Bachelor's Thesis: Domain-Adversarial Transfer Learning Spring 2024

- Designed and implemented an intermediate-domain transfer learning method built on Domain-Adversarial Neural Networks (DANN) to reduce source-target feature distribution mismatch.
- Constructed intermediate domains via Mixup/CutMixup between matched source/target classes, enabling a staged adaptation path (source → intermediate → target) in PyTorch.
- Evaluated on Office-31, ImageCLEF, and OfficeHome; intermediate domains improved average accuracy from 76.84% to 95.00% on ImageCLEF (+18.16 pp) and from 65.00% to 67.06% on OfficeHome (+2.06 pp), while maintaining 100% on Office-31.
- GitHub Link

MUSIC TECHNOLOGY WORKS COLLECTIONS

Audio Visualizer JUCE Plugin

December 2025

- Audio Visualizer is a real-time audiovisual plugin built with JUCE, translating live audio into an expressive, physically inspired particle system inside a DAW. The plugin runs entirely in real time, analyzing incoming audio on the audio thread and mapping continuous control signals to a custom visual engine that reacts to musical energy and timbral change. Its particle-orbit visual model drives motion, scale, color, and rotation through smoothed audio parameters. An optional Emotion mode uses learned latent controls to shape higher-level visual behaviors, allowing the visualization to feel responsive rather than purely reactive.
- GitHub Link

Monophonic MIDI Synthesizer (Max/MSP)

December 2025

- An interactive and visualized monophonic MIDI synthesizer built in Max/MSP, designed as a playable instrument that supports real-time performance through both a virtual on-screen keyboard and external MIDI controllers. The synthesizer integrates waveform selection, ADSR envelope shaping, LFO-based modulation (frequency and depth), nonlinear overdrive, filtering, and time-based effects, with each processing module individually controllable and bypassable.
- GitHub Link

Interactive Music Generation via Image-Based Audio Parameter Mapping

September 2024

- This project explores real-time music generation driven by image features, using Max/MSP to convert visual information (color, brightness, saturation, and spatial position) into dynamic audio parameters. By continuously analyzing the color block at the user's current mouse position, the system creates an interactive audiovisual environment where "what you see" and where you choose to look directly shapes "what you hear."
- Google Drive Link

RECORDING ENGINEERING & IMMERSIVE AUDIO SESSIONS

NYU Dolan Studios / Paulson Center

Summer 2025

- Participated in professional multi-microphone ensemble recording sessions using Tonmeister and immersive audio recording techniques.
- Worked with stereo, surround, and 3D microphone array configurations including M/S, Hamasaki Square, Ipsi arrays, and Ambisonics-based spatial capture workflows.
- Assisted in microphone placement, session setup, monitoring, and ensemble recording for jazz sextet, brass quartet, wind quintet, percussion ensemble, piano trio, and tango ensemble sessions.
- Gained hands-on experience with professional studio microphones and immersive recording systems including Sennheiser MKH 800 Twin, Ambeo VR, Neumann U87/M147, Coles 4038, and KM184 microphones.
- Contributed to multichannel recording workflows involving spatial imaging, room acoustics, ensemble balance, and immersive audio capture.
- **Selected Sessions:** Brass Quartet (3D), Jazz Sextet (Hamasaki), Wind Quintet, Marimba Quintet, Piano Trio, String Trio, Jazz Organ.

MUSIC WORKS COLLECTIONS

Misty Forest

December 2024

- Live sound processing performance using Max/MSP, integrating pre-recorded clarinet material with real-time MIDI-controlled transformations.
- Designed multi-patch system combining spectral processing (FFT), delay, filtering, and algorithmic control to create an interactive and immersive sonic texture.
- Structured the performance into multiple controllable segments with a text-based score, enabling real-time interaction and adaptive sound evolution during live execution.
- [Google Drive Link](#)

Walking Through the Dark

November 2024

- Algorithmic composition developed in SuperCollider, constructing multiple synthesized sound modules including pads, arpeggios, and noise-based textures.
- Designed a modular synthesis system combining oscillators, frequency modulation, filtering, and envelope control to create evolving and narrative-driven soundscapes.
- [Google Drive Link](#)

Playing Bass in a Mountain

February 2024

- Mixed-media composition combining live bass performance with sound transformation and electronic production in Logic Pro.
- Designed a sound design pipeline transforming bass guitar techniques into pads, percussive elements, and synthesized textures using delay, reverb, and granular effects.
- Structured the piece into multiple narrative-driven sections with automated parameter control, creating an evolving and immersive sonic environment.
- [Google Drive Link](#)

Breaking Free

January 2024

- Composed, produced, and performed an original EDM track, integrating MIDI arrangement, live guitar recording, and vocal production across Logic Pro and Ableton Live.

- Designed full production workflow including sound selection, arrangement, vocal recording, and mixing/mastering for a genre-oriented electronic music piece.
- Developed dynamic song structure combining verse, build-up, and drop sections, incorporating layered synthesis and instrumental design to create contrast and energy progression.
- [Google Drive Link](#)

The Evil Within 2 Trailer Soundtrack

October 2023

- Composed and produced a full trailer soundtrack for a psychological horror game, designing a multi-section structure (intro, bridge, climax, ending) synchronized to visual narrative.
- Integrated sound design and music composition, combining synthesized textures, vocal recording, and environmental sound effects to create immersive horror atmospheres.
- Applied advanced audio processing techniques including EQ automation, spatial layering, and dynamic arrangement to enhance tension and cinematic impact.
- [Google Drive Link](#)

Dancing in the Rain

April 2023

- Composed an original solo piano piece inspired by rainy-day scenes in Changsha, focusing on emotional expression and atmospheric storytelling.
- Explored melodic phrasing and harmonic progression to evoke a sense of solitude and introspection.
- [NetEase Cloud Music Link](#)

Lonely Travel

September 2022

- Composed and produced an experimental electronic piece in Ableton Live, inspired by real-world narrative scenarios and structured into multi-stage storytelling sections.
- Designed custom synthesizer patches using LFO-controlled modulation and filtering to generate evolving ambient textures and stochastic timbral variations.
- Transformed environmental audio samples into musical elements through audio effects processing (resonator, flanger, chorus), integrating sound design with compositional structure.
- [SoundCloud Link](#)

TECHNICAL SKILLS

1. *Programming*: Python, C/C++ (JUICE), MATLAB, SQL
2. *Machine Learning*: PyTorch, Transformers, Diffusion Models, VAEs, scikit-learn, Representation Learning
3. *Audio & MIR*: Music Information Retrieval, Audio DSP, Machine Listening, Source Separation, Symbolic Music Generation
4. *Systems & Tools*: Linux/Unix, Git, Docker, REST APIs